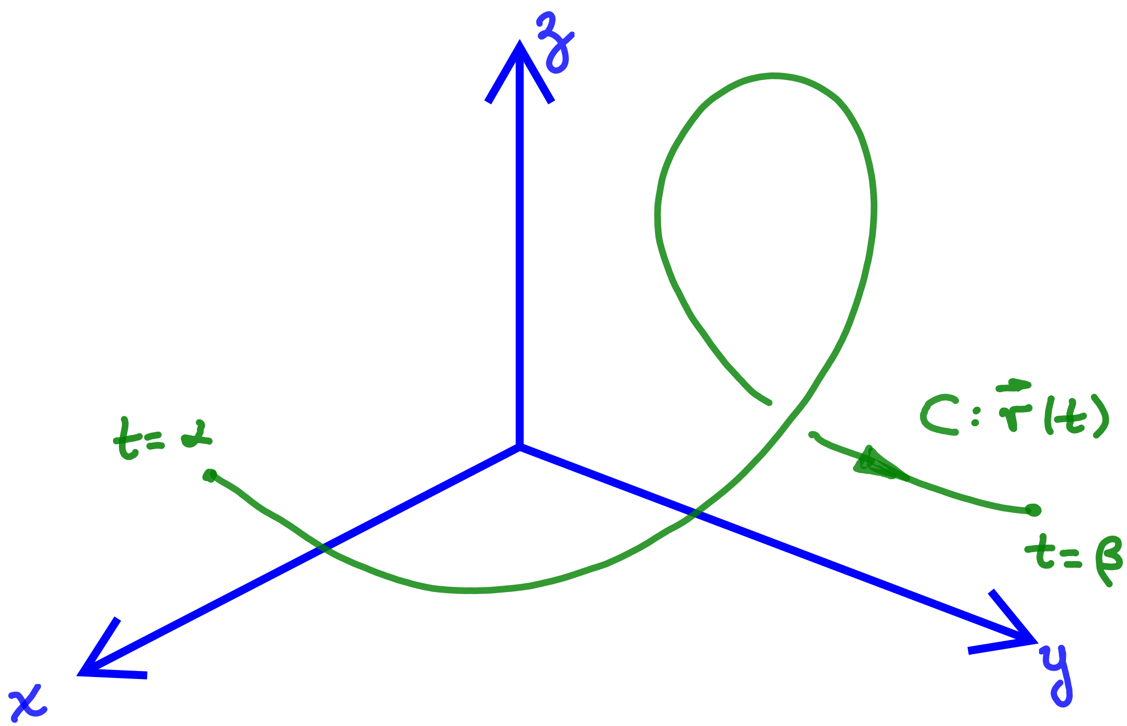
Arc Length, Unit Tangent Vector and Curvature

Trim Ch 11, S. 11.11



Outline:

1. Arc Length Parameter and shape of a curve
2. Unit Tangent Vector and Curvature

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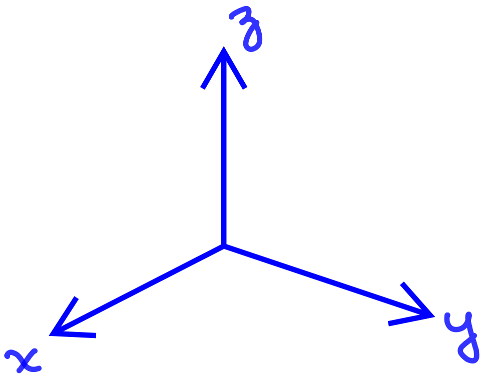
Arc Length Along a 3D Space Curve

One of the features of a smooth curve is that it has a measurable length.

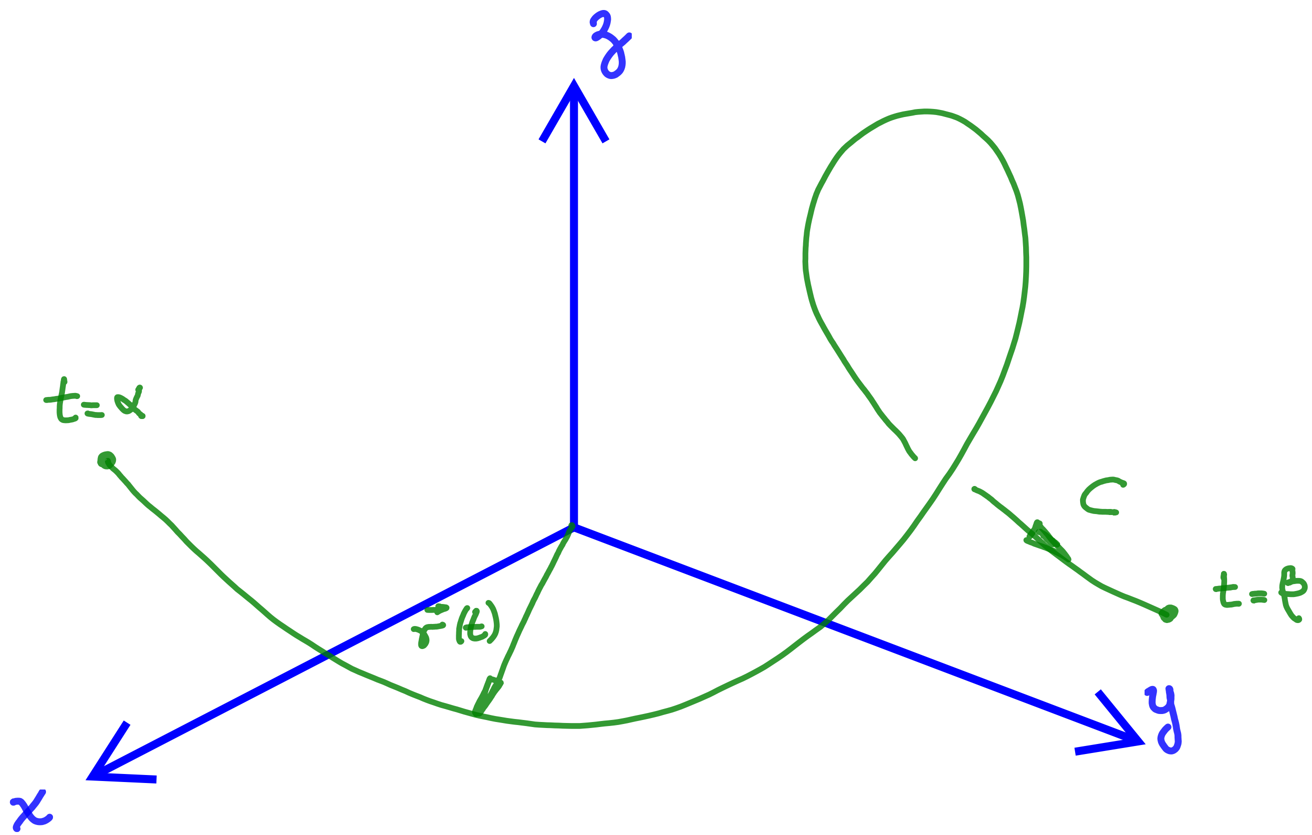


The coordinate of each point "s" is the directed distance along the curve C from a pre-selected base point.

Arc Length Along a 3D Space Curve

Let us consider a smooth curve C defined by the path:

$$\vec{r}(t) = x(t)\hat{\imath} + y(t)\hat{\jmath} + z(t)\hat{k}, \quad \alpha \leq t \leq \beta$$



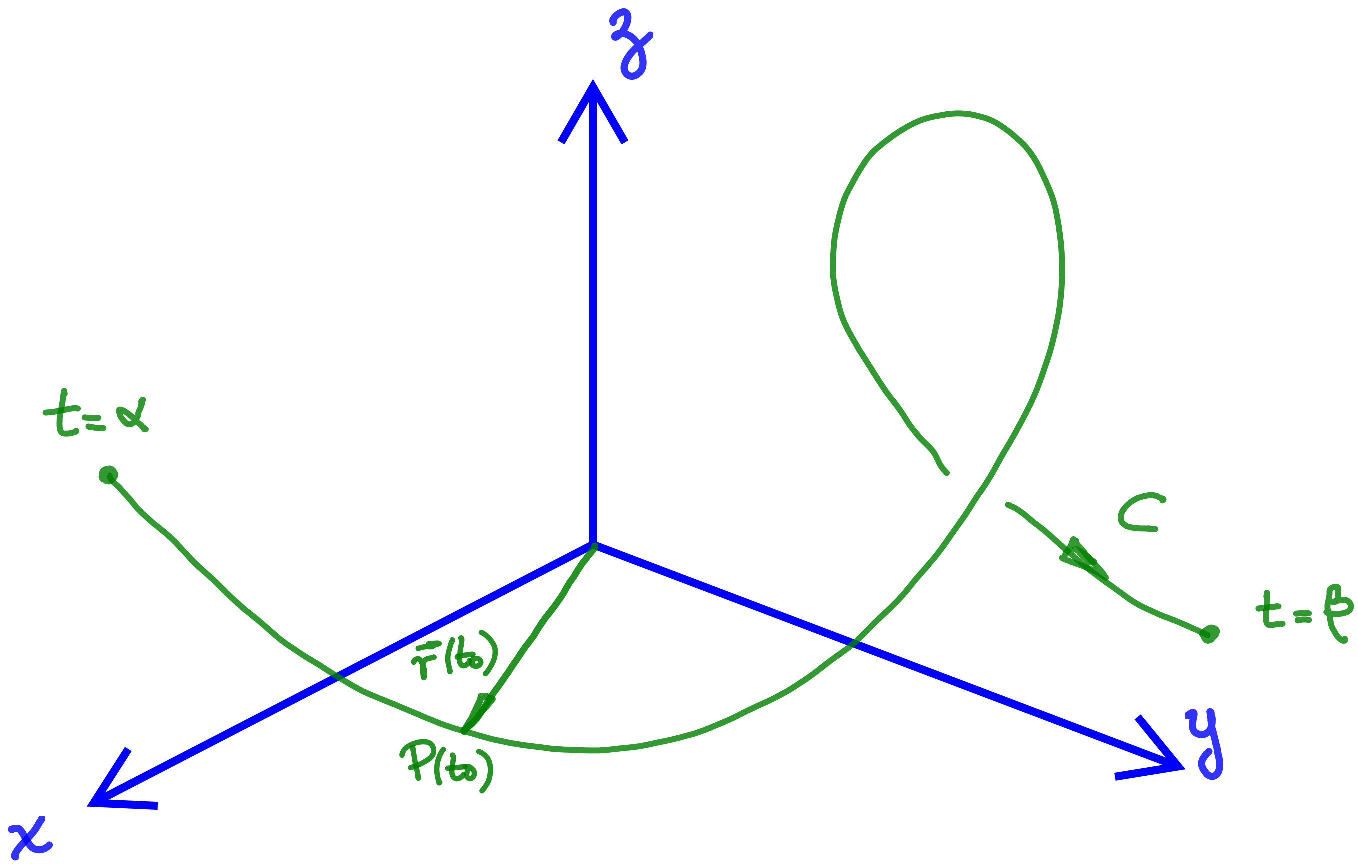
Arc Length Parameter

The parameter "t" is the natural parameter for describing a moving body's velocity and acceleration. The parameter "s" is the natural parameter for studying the shape of a curve and it is called the arc length parameter.

Relating the parameters "s" and "t":

Arc Length Parameter

Arc Length Function



We call the directed distance $s(t)$ the arc length function the curve C

Arc Length Function vs. Arc Length Parameter

The arc length function can be calculated as

Each value of the arc length function $s(t)$ determines a point in C so we can write,

$\vec{r}(s)$ parametrizes C with respect to the arc length parameter "s", and it determines where we are on the curve after traveling a distance "s"

Example

Reparametrize the path $\vec{r}(t)$ below with respect to arc length, measured from the point where $t = 0$ in the direction of increasing t.

$$\vec{r}(t) = (1 + 2t)\hat{\imath} + (3 + t)\hat{\jmath} - 5t\hat{k}$$

Solution

Example

The "s" parametrization is generally difficult to find analytically for a curve that is already given in terms of the parameter t. Fortunately, we rarely need an exact formula for $s(t)$ or its inverse $t(s)$.

See **video posted in LEARN** for a numerical example of arc length parametrization of a curve C initially parametrized by "t"

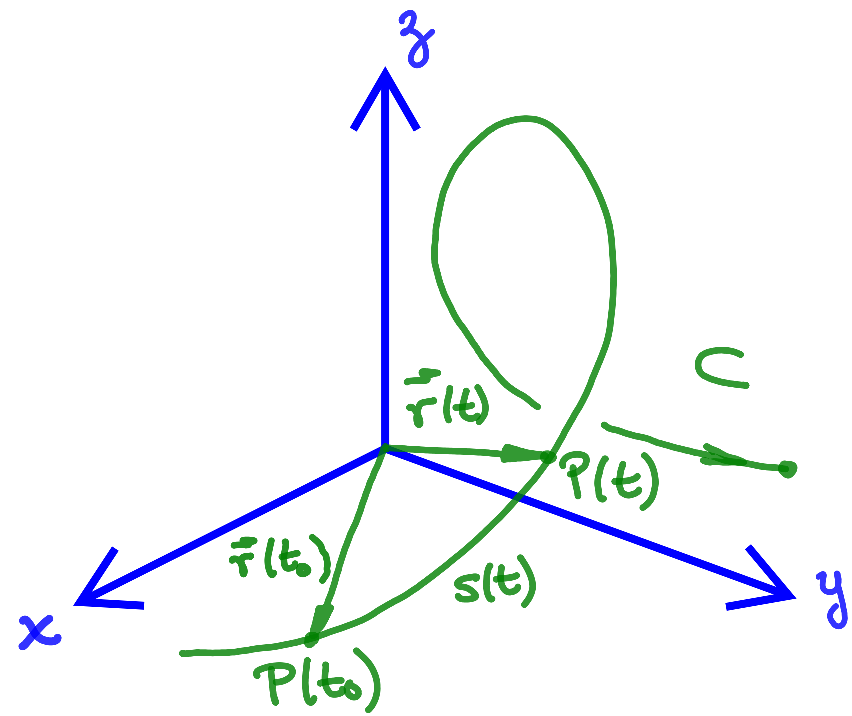
How does arc length parameter increase?

If $t > t_0$, $s(t)$ is the distance from P_0 to P,

$$s(t) = \int_{t_0}^{t} |\vec{v}(\tau)| d\tau$$

and $s(t)$ increases with "t"

If $t < t_0$, $s(t)$ is the distance from P to P_0,



Total Length of a Curve

For a smooth curve C defined by the path:

$$\vec{r}(t) = x(t)\hat{\imath} + y(t)\hat{\jmath} + z(t)\hat{k}, \quad \alpha \le t \le \beta$$

The total length of the curve C from α to β is,

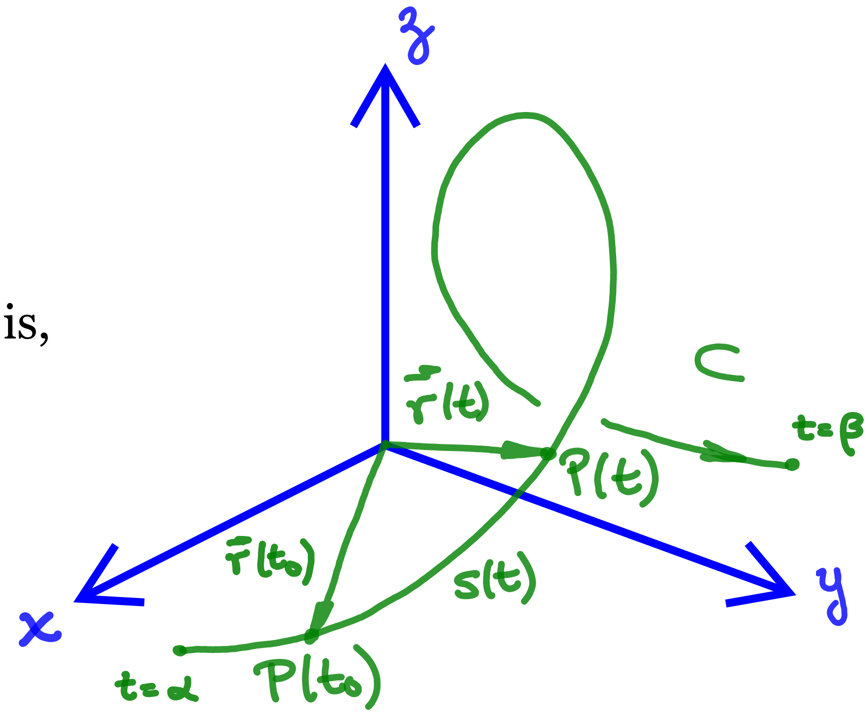
$$L = \int_{\alpha}^{\beta} |\vec{v}(t)| dt$$

Since,

$$\vec{v}(t) = \frac{dx(t)}{dt}\hat{\imath} + \frac{dy(t)}{dt}\hat{\jmath} + \frac{dz(t)}{dt}\hat{k}$$

Then,

$$L = \int_{\alpha}^{\beta} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}\, dt$$



3D Version of what you learned before!

So far …

- The arc length parameter is used to mathematically describe the shape of a curve with a new parametrization $\vec{r}(s(t))$.
- The parametrization $\vec{r}(t)$ determines where we are on the curve C after a time "t".
- The parametrization $\vec{r}(s)$ determines where we are on the curve C after traveling a distance "s".
- Both "t" and "s" increase in the direction of motion.
- The arc length parameter is used to determine the total length of a curve described by $\vec{r}(t)$, where $\alpha \leq t \leq \beta$.

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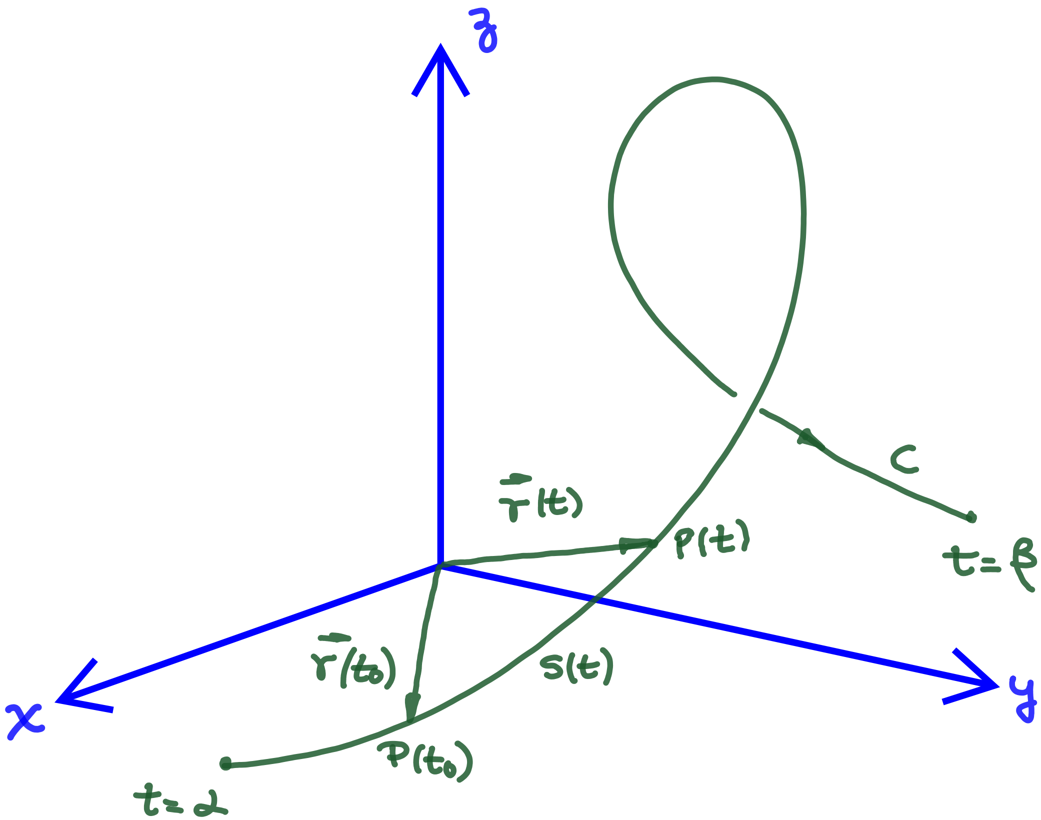
Unit Tangent Vector



For our "t" parametrization we found that the rate of change $\frac{d\vec{r}}{dt} = \vec{v}(t)$ represents the turning tangent to the curve C at P

The turning **unit tangent vector** $\hat{T}(t)$ to the curve C is therefore given by,

Unit Tangent Vector

Now, looking at our "s" parametrization, what does $\frac{d\vec{r}}{ds}$ mean?

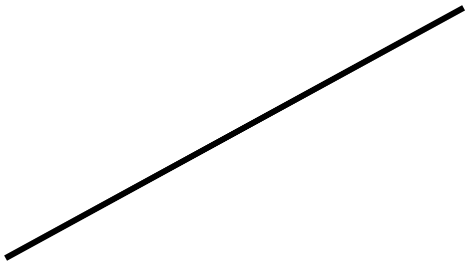
Curvature

The rate at which $\hat{T}$ turns per unit length along the curve is called the curvature, which is represented by κ.

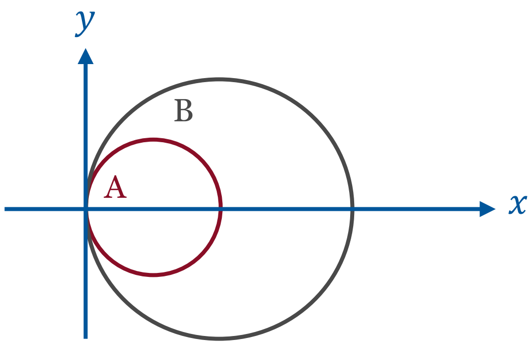
For a smooth curve C parametrized in terms of "t" (that is , $\vec{r}(t)$), we can calculate the curvature as a function of "t" using the chain rule of differentiation,

Curvature of some familiar shapes

1. Straight Line



2. Circles



Summary

- The parameter "s" is the natural parameter to study the shape of a curve
- We used "s" to determine the length of a curve in 3D
- The unit tangent vector $\hat{T}$ turns as P moves along the curve in the direction of increasing arc length "s"
- The curvature is the rate of change of $\hat{T}$ with respect to "s"

Thank you